



WSQ COURSE SLIDES

Data Visualisation with Tableau

Course Code: TGS-2020503177

Data Visualization · ICT-DIT-3006-1.1

Trainer: Dr Alfred Ang

Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v17.0 · 15 August 2026



Concept visual generated for this course

 COURSE ADMINISTRATION

Welcome and Course Orientation

Digital Attendance (Mandatory)

1

AM attendance

Complete SSG digital attendance at the start of each day.

2

PM attendance

Complete the second attendance at the instructed time.

3

Assessment attendance

Take the separate digital attendance before the papers.

4

Eligibility

Maintain at least 75% attendance and pass the assessment.

About the Trainer



General Trainer

Complete this profile before delivery

NAME

TITLE / DESIGNATION

QUALIFICATIONS

EXPERTISE

EXPERIENCE

CONTACT

About the Trainer



Dr Alfred Ang

Principal Trainer · Tertiary Infotech
Academy

ROLE

Principal Trainer and Chief Instructional Designer

EXPERTISE

Data visualisation, analytics, AI and business reporting

CERTIFICATION

ACLP-certified adult educator

DELIVERS

WSQ and professional technology programmes

EXPERIENCE

Cross-industry analytics and workforce training

CONTACT

enquiry@tertiaryinfotech.com

Let's Know Each Other

1

Your role

What decisions do you support?

2

Your Tableau baseline

What have you already built?

3

Your data challenge

Where does interpretation break down?

4

Your outcome

What visual decision product do you want to create?

Ground Rules

1

Be present

Return on time and complete digital attendance.

2

Participate

Ask, test and explain your reasoning.

3

Respect

Critique ideas, not people.

4

Protect data

Use only the supplied synthetic datasets.

5

Save versions

Keep source, work-in-progress and packaged evidence.

6

Learn by doing

Attempt before opening the sample workbook.

Lesson Plan — Two Days

DAY 1

- 1 9:00–10:30 · Tableau foundations
- 2 10:40–12:30 · Labs 1–2
- 3 1:30–3:15 · Sources and chart choice
- 4 3:25–6:00 · Labs 3–5

DAY 2

- 1 9:00–10:30 · Dashboards and actions
- 2 10:40–12:30 · Labs 6–8
- 3 1:30–4:00 · Calculations, analytics and Labs 9–10
- 4 4:00–6:00 · WA and PP assessment

Learning Outcomes

1

LO1

Apply basic Tableau features for displaying information.

2

LO2

Synthesise various plots to present data visually.

3

LO3

Transform data to create informative and dynamic data displays.

4

LO4

Create dashboards and scorecards to display internal and external benchmark data.

5

LO5

Apply calculations and parameters to create interactive graphics, visuals and technical features.

6

LO6

Perform analytics and communicate data limitations and interpretations of findings.

Skills Framework Alignment

1

K1

Interpretation of data analysis and findings

2

K2

Types of information displays

3

K3

Suitability of data representations and visual displays for different contexts

4

K4

Data visualisation tools and techniques

5

K5

Elements of data dashboards

6

A1–A6

Practical abilities span technique selection, visual organisation, dynamic displays, dashboards, interactivity and responsible interpretation.

Briefing for Assessment

1

Individual

No discussion or sharing during either paper.

2

Open book

Use slides, Learner Guide and approved materials only.

3

Devices

Place phones and unrelated materials under the table.

4

Evidence

Submit your own Tableau workbook and screenshots.

5

Integrity

No photos, recording or verbatim copying.

6

Process

Re-assessment and appeal arrangements are available.

Assessment and Funding

Written Assessment

- 5 open-ended questions
- 60 minutes · open book

Practical Performance

- 6 connected Tableau tasks
- 60 minutes · open book

Outcome

- Competent or Not Yet Competent
- Attendance and submission requirements apply

Assessment Flow



Access the Hands-On Labs

Option A · Clone

- Clone the GitHub repository
- Open the numbered lab folder
- Start with the root Excel file

Option B · Download ZIP

- GitHub > Code > Download ZIP
- Extract the full folder tree
- Keep Tableau and Excel together

Open GitHub repository

Lab discipline

- Use the LG for detailed procedure
- Reconcile against Excel controls
- Package evidence as .twbx

Download Course Material

1 · Sign in

- Open the LMS/TMS
- Use your learner account
- Select this course

2 · Download

- Learner slides and guide
- Keep the lab folders intact
- Use the current version

3 · Verify

- Check TGS-2020503177
- Confirm version 17.0
- Ask the trainer if files differ

Open courseware portal

Tableau Safety and File Discipline

1

Public platform

Tableau Public is public by design; use only supplied synthetic data.

2

Packaged evidence

Save learner work as .twbx so local files travel with the workbook.

3

Source preservation

Keep the supplied Excel source unchanged and reconcile against its Control sheet.

4

Responsible claims

State filter scope, assumptions, limitations and monitoring trigger.

From Source to Decision Story



LEARNING JOURNEY · 6 TOPICS

Connect → Model → Visualise → Interact → Analyse → Communicate

The six linked topics progress from reliable source connection to responsible decision communication; detailed procedures stay in the Learner Guide.

TOPIC 01

Basic Tableau Features

Tableau ecosystem · interface · data roles · shelves · hierarchy

Key Concepts — Basic Tableau Features

1

Visual analysis cycle

Connect, explore, explain and refine around a decision question.

2

Dimensions and measures

Dimensions slice the view; measures are aggregated to quantify it.

3

Discrete and continuous

Blue headers create categories; green axes create a continuous range.

4

Workbook structure

Worksheets build views; dashboards combine them; stories sequence insights.

5

Fields and metadata

Names, types, roles, aliases and default properties shape every view.

6

Hierarchy

Ordered levels let users drill from overview to operational detail.

What Tableau Adds to Decision Work

Fast exploration

- Drag-and-drop visual analysis
- Immediate feedback on questions

Connected evidence

- Files, databases and governed sources
- Live or extracted data

Communicable output

- Worksheets, dashboards and stories
- Publish with context and controls

The Visual Analysis Cycle

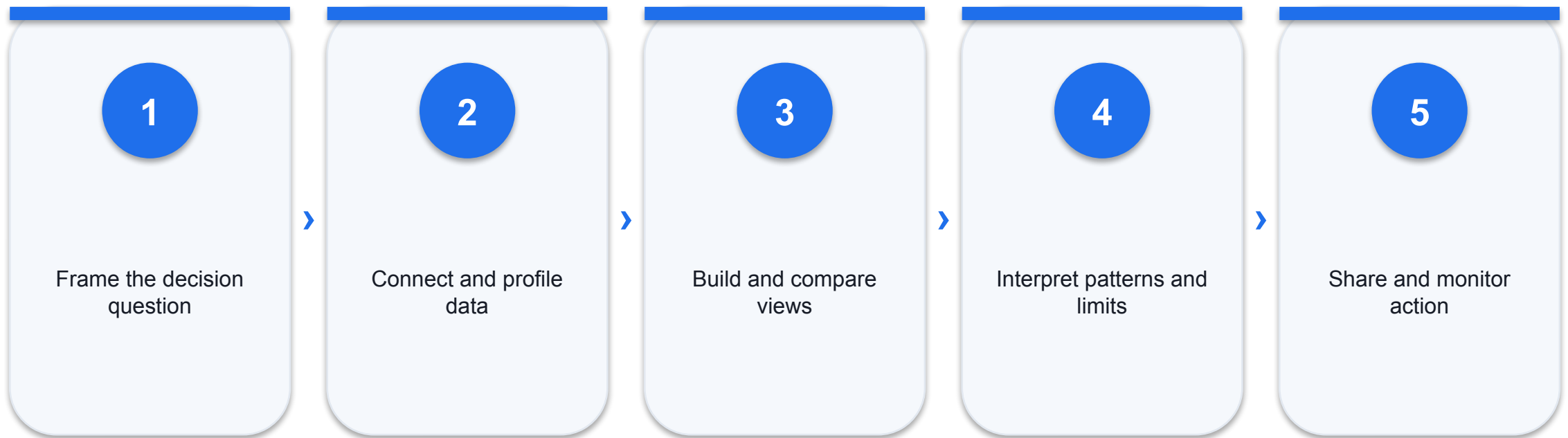


Tableau Products and Roles

CONCEPT

- 1 Desktop / Public author views and workbooks.
- 2 Cloud / Server govern sharing and collaboration.
- 3 Prep supports repeatable cleaning and reshaping.
- 4 Reader opens packaged workbooks offline.

DECISION LENS

- 1 Choose by data sensitivity and audience.
- 2 Tableau Public is public by design.
- 3 Publishing permissions are part of analysis quality.
- 4 The course uses Desktop or Public for authoring.

Workspace: One Canvas, Several Control Surfaces

1

Data pane

Fields, folders, hierarchies and metadata.

2

Shelves

Rows, Columns, Pages and Filters shape structure.

3

Marks card

Colour, size, label, detail, tooltip and shape.

4

View

Marks, axes, headers, legends and annotations.

Dimensions and Measures

DIMENSIONS

- 1 Dimensions describe who, what, where and when.
- 2 They create headers and define the level of detail.
- 3 Examples: Region, Category, Customer and Order Date.

MEASURES

- 1 Measures quantify how much or how many.
- 2 They are usually aggregated in the view.
- 3 Examples: Sales, Profit, Quantity and Resolution Hours.

Discrete and Continuous

1

Discrete field

Creates separate headers; shown as a blue pill.

2

Continuous field

Creates an axis across a range; shown as a green pill.

3

Date choice

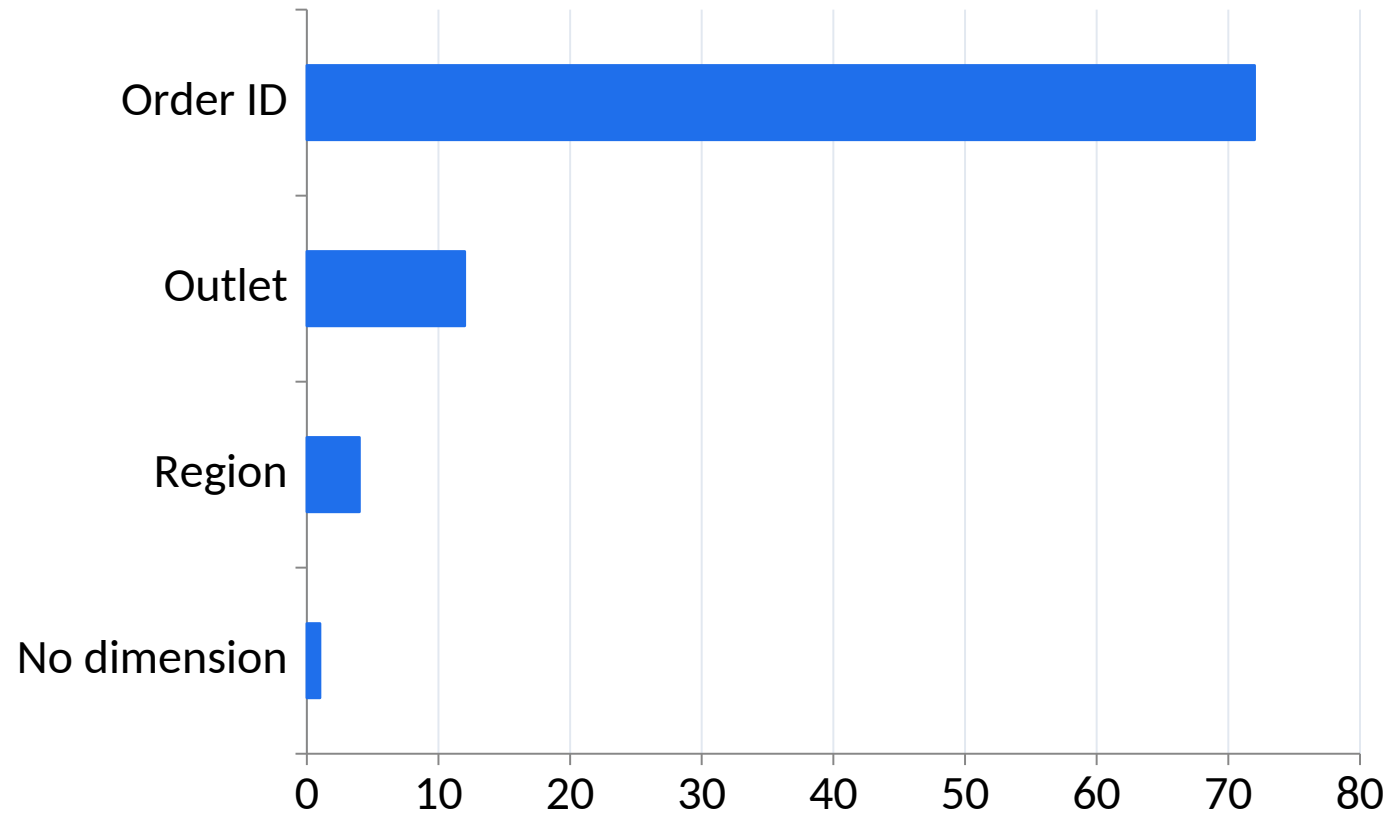
Year may be discrete while exact date can be continuous.

4

Design effect

The same source field can create a different visual structure.

Measures Aggregate at the View Level



READ THE EVIDENCE

Adding dimensions increases the number of marks and changes the level at which a measure is aggregated.

Editable chart: change the embedded data, labels, axes and colours directly in PowerPoint.

Rows, Columns and the Level of Detail

1

Rows

Positions headers or axes vertically.

2

Columns

Positions headers or axes horizontally.

3

Detail

Adds marks without necessarily adding a visible encoding.

4

Granularity

Every added dimension can change the number of marks.

Marks Encode Meaning

Position and length

- Strong for accurate comparison
- Preferred for core quantitative questions

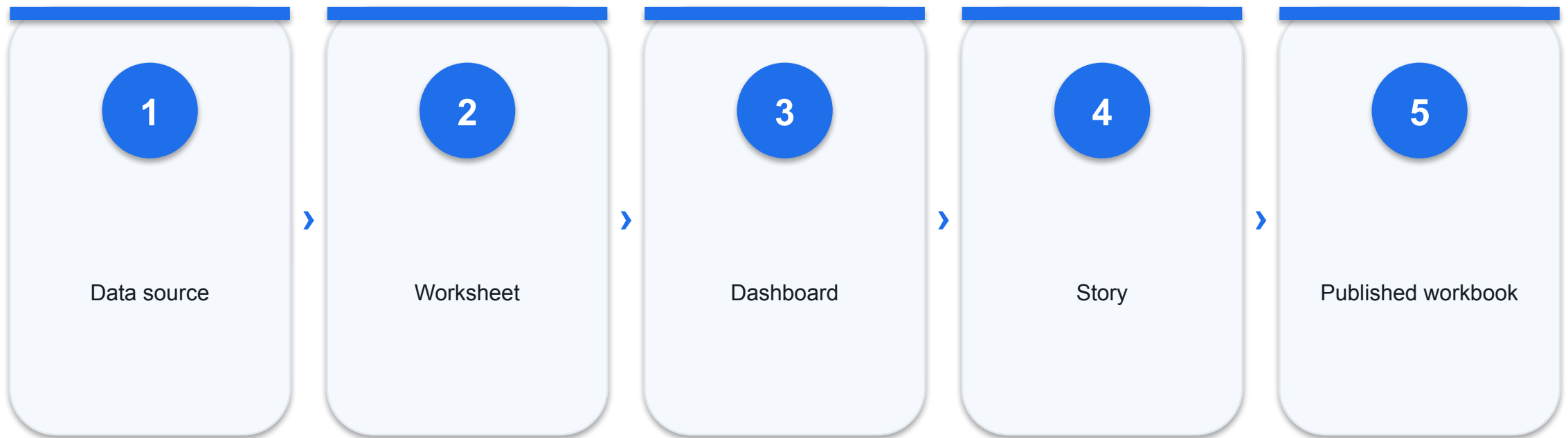
Colour and size

- Useful for emphasis and additional variables
- Need accessible legends and restraint

Label and tooltip

- Add exact values and context
- Should not compensate for a weak view

Workbook Structure



File Types and Portability

1

.twb

XML workbook definition; external data remains separate.

2

.twbx

Packaged workbook with local data and assets.

3

.hyper

Tableau extract optimised for analytical queries.

4

.tds / .tdsx

Reusable data-source metadata, optionally packaged.

Folders and Hierarchies

FOLDERS

- 1 Folders organise the authoring experience.
- 2 They do not change the source data or aggregation.
- 3 Use names that reflect business domains.

HIERARCHIES

- 1 Hierarchies define an ordered drill path.
- 2 Typical paths: Region > District > Outlet.
- 3 Every level should answer a meaningful next question.

Connect and Profile Service Operations Data

LAB 1

A customer-experience manager needs a first reliable view of service tickets across channels and regions.

EXPECTED OUTPUT

A connected Tableau workbook with validated field roles and a service-volume view by region and channel.

TOOLS Tableau Desktop/Public · Data Source page · Dimensions/Measures · Show Me

Lab 1 Acceptance Evidence

ACCEPTANCE TEST

The workbook connects without broken fields; dates and numbers have correct roles; the view reconciles to the Excel record count.

Build

Packaged Tableau workbook
Required views and controls

Prove

Screenshot with filter state
Excel reconciliation check

Explain

Observation and meaning
Limitation and next action

Build Hierarchies and Drill Paths

LAB 2

Retail leaders need to move from national performance to region, outlet and department detail without losing context.

EXPECTED OUTPUT

A geographic and product hierarchy with a drillable sales-and-margin view.

TOOLS Folders · Hierarchies · Drill up/down · aliases · default properties

Lab 2 Acceptance Evidence

ACCEPTANCE TEST

Both hierarchies drill in the correct order and totals remain stable at every displayed level.

Build

Packaged Tableau workbook
Required views and controls

Prove

Screenshot with filter state
Excel reconciliation check

Explain

Observation and meaning
Limitation and next action

Recap — Basic Tableau Features

1

Knowledge

Visual analysis cycle and related principles.

2

Ability

K1 · A1 · LO1

3

Evidence

A packaged Tableau workbook plus reconciliation and interpretation.

4

Next

Connect this layer to the next decision requirement.

TOPIC 02

Data Sources and Visualisation

Connections · extracts · relationships · joins · blending · chart choice

Key Concepts — Data Sources and Visualisation

1

Connection strategy

Choose live or extract by freshness, performance and governance needs.

2

Relationships

Preserve each table's grain and combine data at analysis time.

3

Joins

Physically combine rows; cardinality and unmatched keys determine the result.

4

Visual encoding

Position and length usually support more accurate comparisons than area or colour.

5

Chart purpose

Use bars for comparison, lines for time, scatterplots for relationship and treemaps for hierarchy.

6

Aggregation

SUM, AVG, COUNT and distinct count answer different business questions.

Live Connection or Extract?

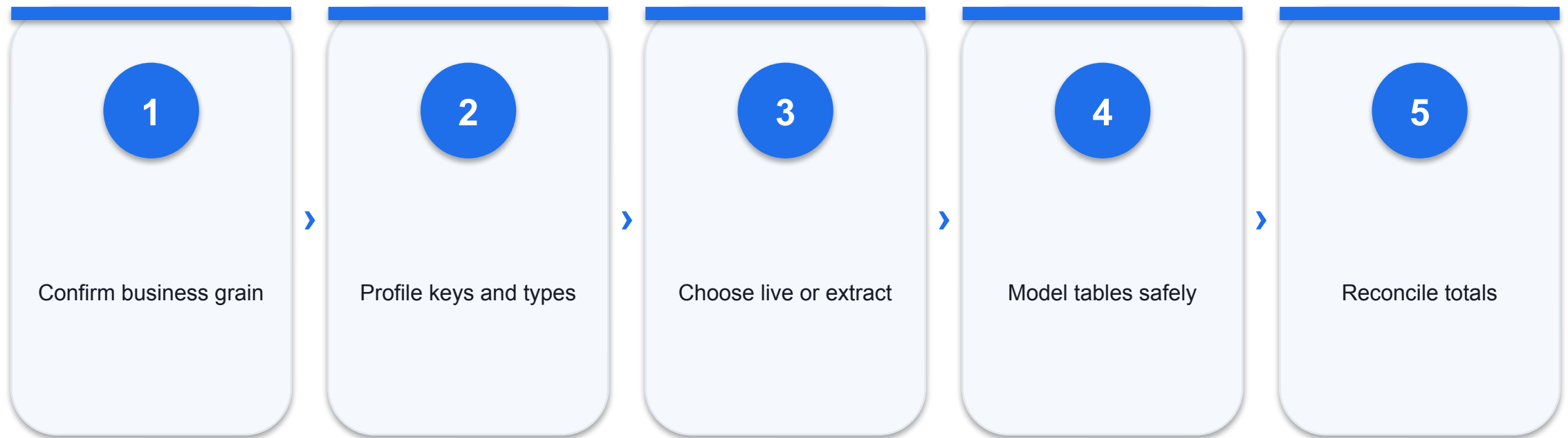
LIVE

- 1 Queries the source at interaction time.
- 2 Best when current values and central controls matter.
- 3 Performance depends on the source and network.

EXTRACT

- 1 Stores a local analytical snapshot.
- 2 Supports fast interaction and offline work.
- 3 Requires a refresh policy and timestamp.

Connection Readiness Gate



Three Ways to Combine Data

Relationships

- Logical layer
- Preserve table grain

Joins

- Physical row combination
- Cardinality changes rows

Blending

- Separate data sources
- Aggregate secondary results

Join Types and Their Consequences

1

Inner

Keeps only matched rows.

2

Left

Keeps every left row plus matching right data.

3

Right

Keeps every right row plus matching left data.

4

Full outer

Keeps matched and unmatched rows from both sides.

Relationships versus Joins

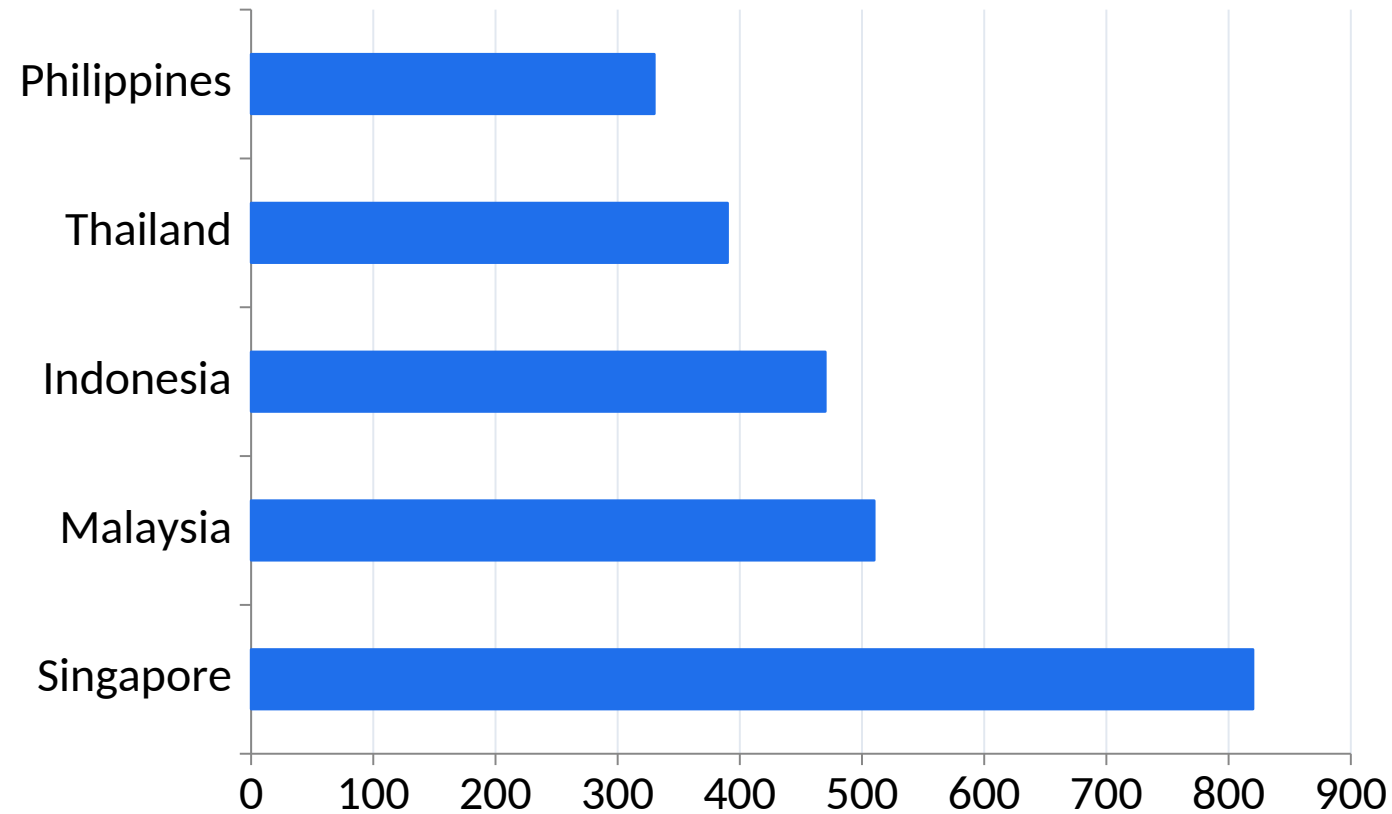
LOGICAL MODEL

- 1 Relationships are resolved in the context of the view.
- 2 They reduce accidental duplication across different grains.
- 3 Use defined cardinality and referential integrity when known.

PHYSICAL MODEL

- 1 Joins create one physical row set before analysis.
- 2 One-to-many matches can duplicate measures.
- 3 Use joins when row-level combination is truly required.

Ranked Comparison: Sales by Market

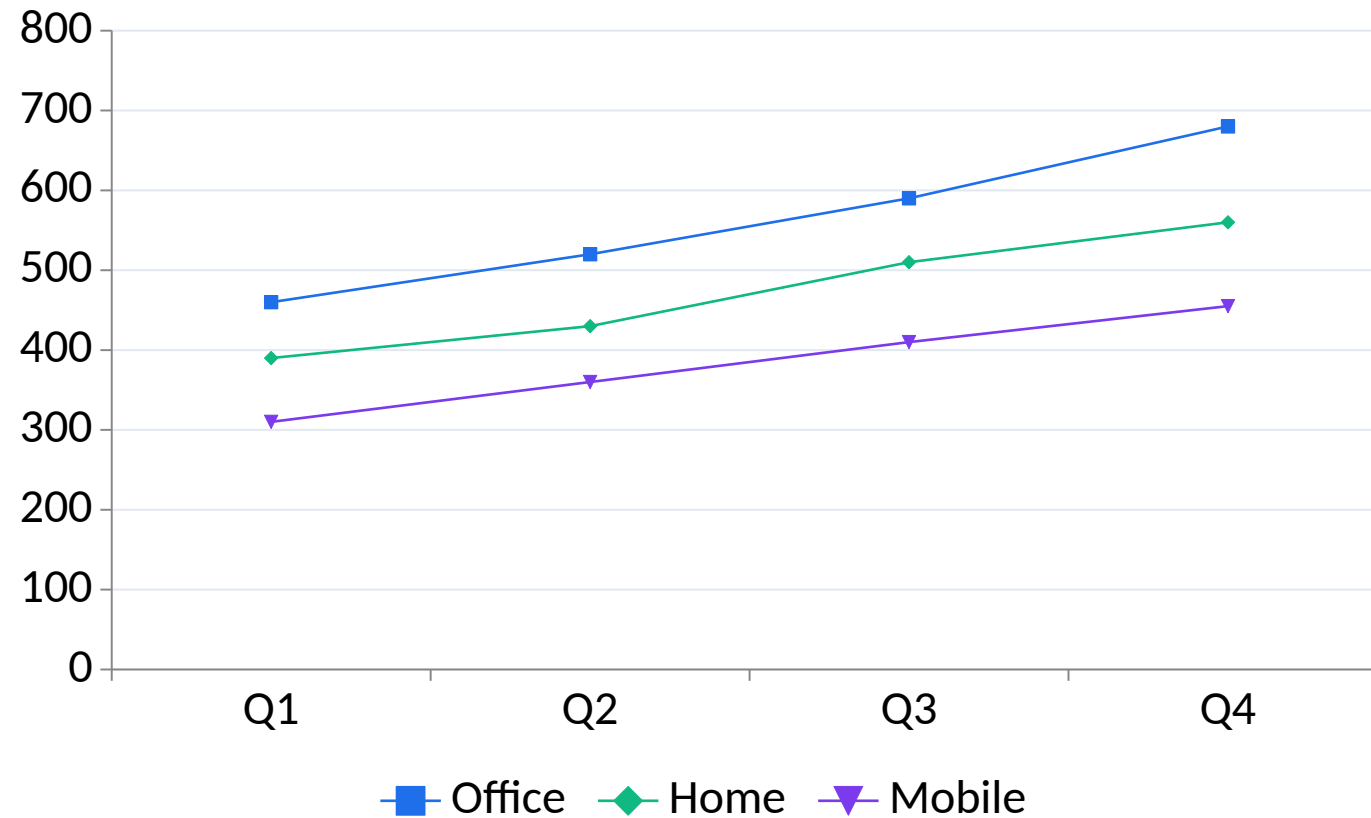


READ THE EVIDENCE

Singapore leads the sample, but regional budget allocation should also consider margin and growth.

Editable chart: change the embedded data, labels, axes and colours directly in PowerPoint.

Trend: Quarterly Sales by Category



READ THE EVIDENCE

**All categories rise in Q4;
Office contributes the largest
absolute increase.**

Editable chart: change the embedded data, labels, axes and colours directly in PowerPoint.

Relationship: Sales and Profit

Pattern

- Marks generally rise from lower-left to upper-right
- Higher sales are associated with higher profit in the sample

Exception

- One high-sales outlet has weaker profit
- Use Outlet on Detail to identify it

Caution

- Association does not establish causality
- Inspect discount, mix, cost and data quality

Choose a Chart from the Question

1

Comparison

Bar or dot plot.

2

Time

Line or area chart with an ordered axis.

3

Relationship

Scatterplot with a meaningful level of detail.

4

Distribution

Histogram or box plot.

5

Composition

Stacked bar or treemap when part-to-whole matters.

6

Location

Map only when geography changes the decision.

Portfolio Views

Treemap

- Hierarchical part-to-whole
- Area is less precise than length

Heatmap

- Pattern across two categorical axes
- Colour scale needs clear meaning

Box plot

- Median, spread and unusual values
- Good for group distributions

Aggregation Changes the Question

CALCULATION

- 1 SUM answers total contribution.
- 2 AVG answers the typical row at the current grain.
- 3 COUNT counts records; COUNTD counts unique entities.
- 4 ATTR signals whether a single value is present.

CONTROL

- 1 Confirm the denominator behind every average.
- 2 Do not average percentages when a weighted ratio is needed.
- 3 Display the level of detail in titles or captions.
- 4 Reconcile to a known control total.

A view is only as trustworthy as its grain.

Before formatting a chart, prove what one row means and whether combining tables changes the population.

Relate Orders, Products and Returns

LAB 3

Meridian Retail SG stores orders, product master data, outlets and returns at different grains.

EXPECTED OUTPUT

A governed multi-table data model plus an extract and reconciliation sheet.

TOOLS Relationships · joins · cardinality · referential integrity · extract

Lab 3 Acceptance Evidence

ACCEPTANCE TEST

Order totals are not duplicated, unmatched keys are visible, and the extract refreshes from the supplied workbook.

Build

Packaged Tableau workbook
Required views and controls

Prove

Screenshot with filter state
Excel reconciliation check

Explain

Observation and meaning
Limitation and next action

Choose and Build Decision Charts

LAB 4

A commercial director needs comparison, trend, relationship and portfolio views for one monthly review.

EXPECTED OUTPUT

Four fit-for-purpose worksheets: ranked bar, quarterly line, profit-sales scatterplot and category treemap.

TOOLS Bar · line · scatterplot · treemap · labels · tooltips · sort

Lab 4 Acceptance Evidence

ACCEPTANCE TEST

Every chart answers its stated question, uses truthful axes, and includes enough context to interpret the marks.

Build

Packaged Tableau workbook
Required views and controls

Prove

Screenshot with filter state
Excel reconciliation check

Explain

Observation and meaning
Limitation and next action

Recap — Data Sources and Visualisation

1

Knowledge

Connection strategy and related principles.

2

Ability

K3 · A2 · LO2

3

Evidence

A packaged Tableau workbook plus reconciliation and interpretation.

4

Next

Connect this layer to the next decision requirement.

TOPIC 03

Data Transformation

Data Interpreter · split · merge · pivot · filters · groups · sets

Key Concepts — Data Transformation

1

Analysis-ready structure

One row is one observation; one column is one variable; one table is one grain.

2

Data Interpreter

Removes presentation clutter from spreadsheets while preserving the underlying evidence.

3

Split and pivot

Separate compound fields and reshape wide columns into analysis-ready rows.

4

Filter order

Extract, source, context, dimension and measure filters do not behave identically.

5

Groups

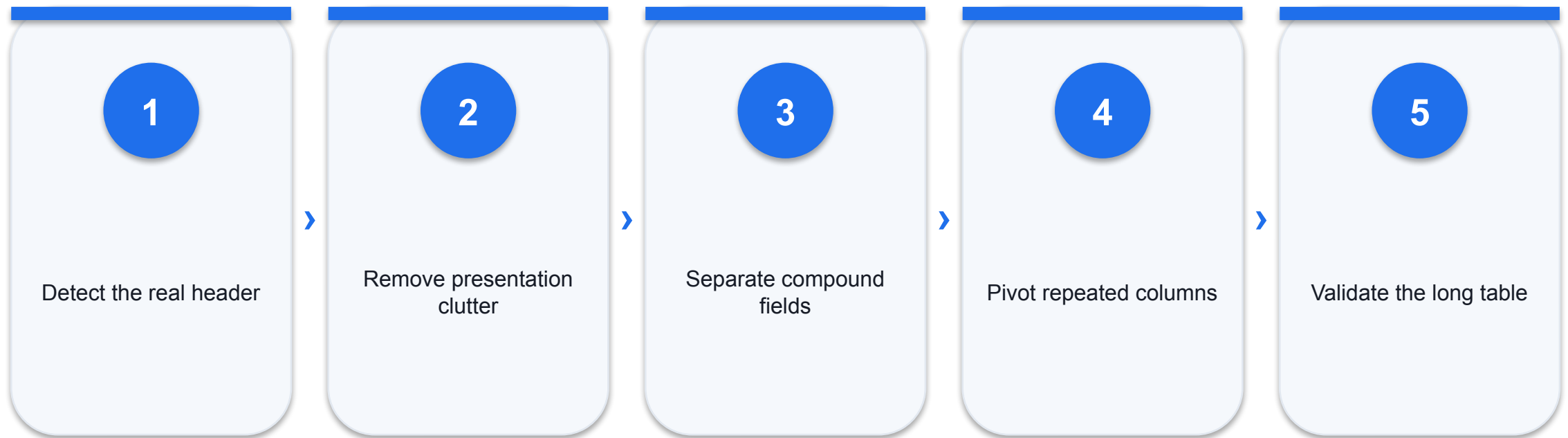
Combine members into a stable business category.

6

Sets

Define a dynamic in/out population that can drive comparisons and actions.

From Presentation Sheet to Analytical Table



What Data Interpreter Can and Cannot Do

AUTOMATION

- 1 Detects headers, notes, blank rows and repeated formatting.
- 2 Helps expose a clean table from a human-formatted sheet.
- 3 Shows what was removed for review.

JUDGEMENT

- 1 It cannot decide the business grain.
- 2 It does not fix ambiguous definitions.
- 3 A clean-looking result still needs reconciliation.

Split, Merge and Aliases

1

Split

Turn Region|District into separate fields.

2

Custom split

Choose delimiter and occurrence deliberately.

3

Merge

Combine fields only when the new meaning is clear.

4

Aliases

Change display labels without changing source values.

Wide versus Long Data

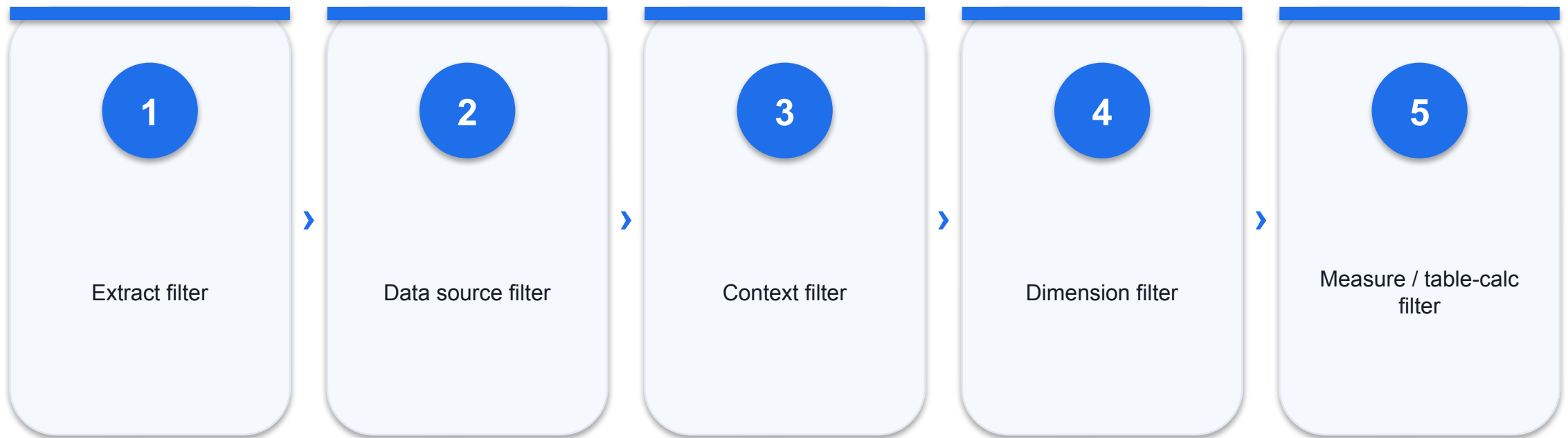
WIDE

- 1 One row may contain Jan, Feb, Mar columns.
- 2 Easy for manual reading; awkward for time analysis.
- 3 New months require new fields.

LONG

- 1 Month becomes a field and value becomes a measure.
- 2 Works naturally with time axes and filters.
- 3 New months append new rows.

Filter Order of Operations



Dimension Filters

1

General

Select explicit members.

2

Wildcard

Match text patterns.

3

Condition

Retain members that meet a rule.

4

Top

Keep leading members by an aggregated measure.

Measure and Date Filters

1

Range of values

Filter aggregated numeric results.

2

At least / at most

Apply a clear materiality threshold.

3

Relative date

Rolling periods that update with time.

4

Range of dates

Fixed window for controlled comparisons.

Context Filters

PURPOSE

- 1 Create a temporary population for dependent filters.
- 2 Useful when Top N should apply within a selected region.
- 3 Can improve or reduce performance depending on design.

RISK

- 1 Do not use context as a visual-formatting trick.
- 2 Explain the population behind the result.
- 3 Test whether changing context changes set membership.

Groups, Sets and Bins

Group

- Stable business consolidation
- Manual or data-driven membership

Set

- Dynamic IN / OUT population
- Supports comparison and actions

Bin

- Intervals for a continuous measure
- Sensitive to bin width

Set Design Patterns

1

Top N

Leading customers or products within context.

2

Condition

Members above a measure threshold.

3

Combined set

Intersection or union of two populations.

4

Set action

Let dashboard interaction change membership.

Transformation Controls

RECONCILE

- 1 Record count before and after.
- 2 Key completeness and uniqueness.
- 3 Total spend, sales or units.
- 4 Explicit list of excluded records.

GOVERN

- 1 Retain the raw source.
- 2 Document every transformation.
- 3 Separate correction from assumption.
- 4 Make limitations visible to the audience.

Cleaning is a chain of evidence, not a cosmetic exercise.

Every transformation should preserve meaning, expose exceptions and be reproducible on refresh.

Clean, Split and Pivot Campaign Data

LAB 5

A campaign export contains banner rows, compound location codes and monthly measures spread across columns.

EXPECTED OUTPUT

An analysis-ready long table with clean location fields and documented transformation checks.

TOOLS Data Interpreter · custom split · merge · pivot · aliases

Lab 5 Acceptance Evidence

ACCEPTANCE TEST

The transformed data has one row per campaign-month, clean location fields, no blank campaign key and reconciled spend totals.

Build

Packaged Tableau workbook
Required views and controls

Prove

Screenshot with filter state
Excel reconciliation check

Explain

Observation and meaning
Limitation and next action

Filter, Group and Set Customer Segments

LAB 6

The loyalty team wants a stable business grouping plus a dynamic high-value customer cohort.

EXPECTED OUTPUT

A segment view using context filters, grouped categories and a top-customer set.

TOOLS Dimension/measure/date filters · context filter · groups · sets

Lab 6 Acceptance Evidence

ACCEPTANCE TEST

The high-value set responds to the selected period and the business grouping is explainable from the source fields.

Build

Packaged Tableau workbook
Required views and controls

Prove

Screenshot with filter state
Excel reconciliation check

Explain

Observation and meaning
Limitation and next action

Recap — Data Transformation

1

Knowledge

Analysis-ready structure and related principles.

2

Ability

K4 · A3 · LO3

3

Evidence

A packaged Tableau workbook plus reconciliation and interpretation.

4

Next

Connect this layer to the next decision requirement.

Lunch Break

45 minutes

TOPIC 04

Dashboards and Stories

KPI scorecards · layout · actions · device design · story sequence

Key Concepts — Dashboards and Stories

1

Four Ws

Who is the audience, what decision, when is it used and where is action taken?

2

Dashboard hierarchy

Lead with decision KPIs, then trends, drivers and diagnostic detail.

3

Layout containers

Tiled containers create predictable alignment; floating objects need deliberate control.

4

Actions

Filter, highlight, URL, navigation, parameter and set actions create purposeful interaction.

5

Benchmark context

Targets, prior periods and peer ranges turn values into performance signals.

6

Stories

A sequence of story points communicates change, evidence and decision.

The Four Ws of a Good Dashboard

1

Who

Audience, authority and data literacy.

2

What

Decision, question and required action.

3

When

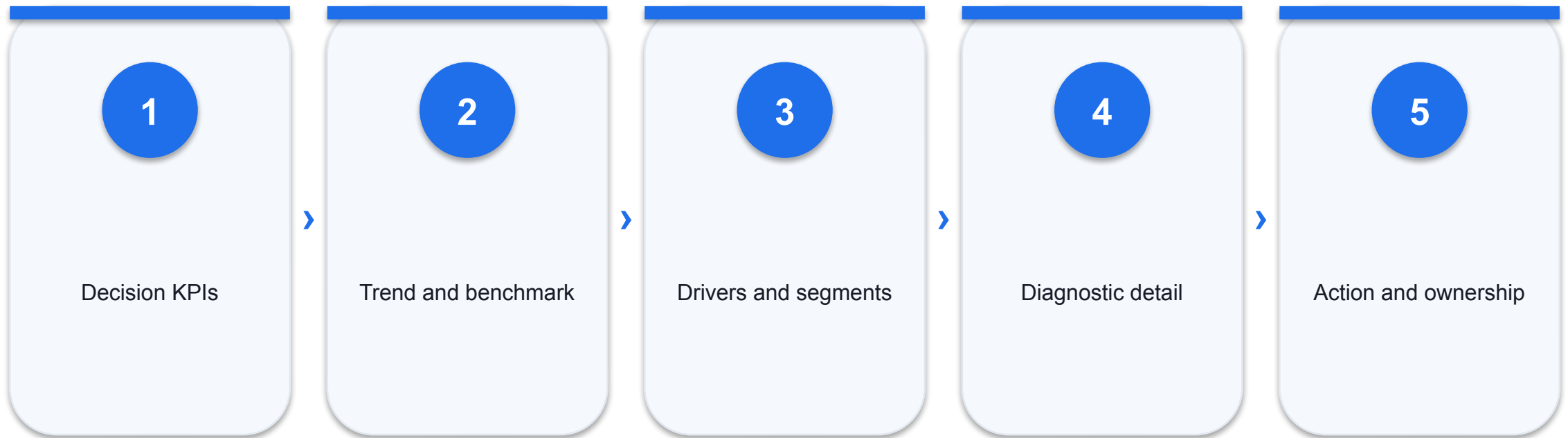
Refresh cadence and time pressure.

4

Where

Presentation, desktop or mobile context.

Dashboard Information Hierarchy



KPI Value versus KPI Context

VALUE

- 1 Value alone states only the current magnitude.
- 2 Large type improves scanning.
- 3 Consistent units and rounding reduce cognitive load.

CONTEXT

- 1 Target, prior period and peer benchmark create meaning.
- 2 Direction should be labelled, not only coloured.
- 3 Variance needs an owner and threshold.

Layout Choices

Tiled

- Predictable alignment
- Responsive container structure

Floating

- Precise placement
- Higher overlap and maintenance risk

Device layout

- Adapt hierarchy to the screen
- Do not merely shrink desktop

Dashboard Objects with a Job

1

Horizontal / vertical container

Structure and alignment.

2

Text

Context, definitions and action.

3

Image

Brand or orientation, used sparingly.

4

Blank

Intentional spacing, not decoration.

5

Web page

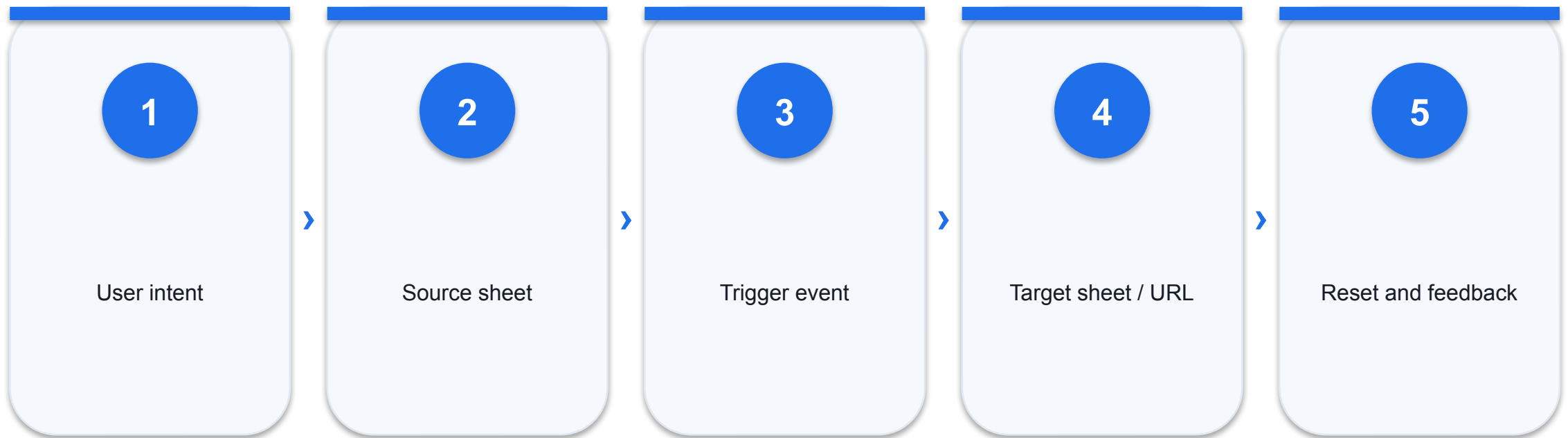
Live reference with security and availability risks.

6

Navigation

Move between summary and detail.

Action Design



Core Dashboard Actions

Filter

- Narrow target views
- Show selected scope

Highlight

- Preserve context
- Emphasise related marks

URL / navigation

- Open supporting detail
- Validate destination and privacy

Advanced Interaction

1

Parameter action

Write a selected value into a parameter.

2

Set action

Change set membership from marks.

3

Change parameter

Switch metric, threshold or view logic.

4

Go to sheet

Move from overview to diagnostic detail.

Dashboard Performance

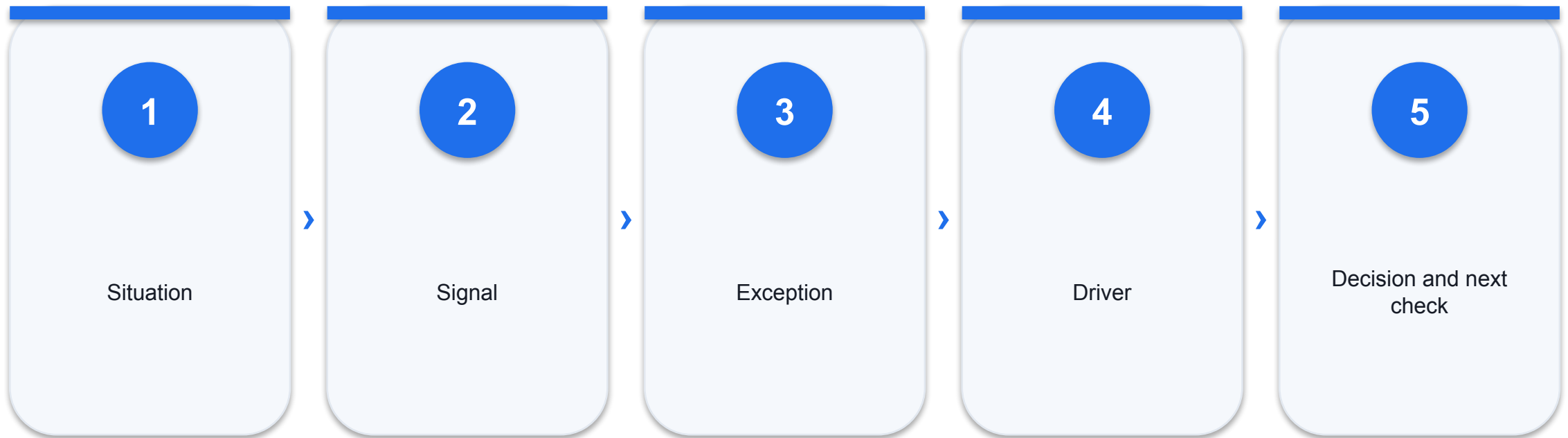
DESIGN

- 1 Limit marks and expensive calculations.
- 2 Use extracts and filters where appropriate.
- 3 Reduce redundant worksheets and high-cardinality quick filters.

TEST

- 1 Measure load and interaction time.
- 2 Performance is part of usability.
- 3 Optimise only after validating the business question.

Story Point Sequence



Dashboard QA

1

Accuracy

Values reconcile to source controls.

2

Clarity

Hierarchy is readable at intended size.

3

Interaction

Actions work and reset predictably.

4

Context

Date, unit, filters and definitions are visible.

5

Accessibility

Colour is not the only cue.

6

Actionability

Owner, threshold and next step are explicit.

Interactivity should reduce uncertainty, not create a puzzle.

Every control needs a clear purpose, visible state and predictable reset path.

Build an Executive Performance Dashboard

LAB 7

Executives need one screen to monitor sales, margin, returns and target attainment across outlets.

EXPECTED OUTPUT

A 16:9 scorecard dashboard with KPI tiles, trend, ranked outlets, benchmark context and visible filter state.

TOOLS Dashboards · layout containers · KPI sheets · device preview · captions

Lab 7 Acceptance Evidence

ACCEPTANCE TEST

The dashboard is readable at presentation size, totals reconcile, active filters are visible and every element supports a named decision.

Build

Packaged Tableau workbook
Required views and controls

Prove

Screenshot with filter state
Excel reconciliation check

Explain

Observation and meaning
Limitation and next action

Add Actions and Tell a Performance Story

LAB 8

Regional managers need to explore a dashboard and then present a concise evidence sequence to leadership.

EXPECTED OUTPUT

A dashboard with filter/highlight/navigation actions and a five-point performance story.

TOOLS Filter action · highlight action · navigation · URL action · story points

Lab 8 Acceptance Evidence

ACCEPTANCE TEST

Actions work from the intended source sheets, preserve context and the story ends with an evidence-backed action.

Build

Packaged Tableau workbook
Required views and controls

Prove

Screenshot with filter state
Excel reconciliation check

Explain

Observation and meaning
Limitation and next action

Recap — Dashboards and Stories

1

Knowledge

Four Ws and related principles.

2

Ability

K5 · A4 · LO4

3

Evidence

A packaged Tableau workbook plus reconciliation and interpretation.

4

Next

Connect this layer to the next decision requirement.

TOPIC 05

Calculations and Parameters

Calculated fields · table calculations · logic · parameters · dynamic views

Key Concepts — Calculations and Parameters

1

Row-level calculation

Evaluated for each record before aggregation.

2

Aggregate calculation

Combines measures at the view's level of detail.

3

Table calculation

Computed over the marks already returned to the visualisation.

4

Logical design

IF, IIF and CASE translate business rules into transparent classifications.

5

Parameters

Single values controlled by the user; pair them with calculations or actions to change behaviour.

6

Dynamic view

A parameter can switch metrics, thresholds, reference values or dimensions.

Three Calculation Levels

Row level

- Evaluated for each source record
- Examples: Sales - Cost

Aggregate

- Evaluated after grouping
- Examples: $\text{SUM}(\text{Profit})/\text{SUM}(\text{Sales})$

Table calculation

- Evaluated over marks in the view
- Examples: percent of total, running sum

Aggregate or Non-Aggregate?

MECHANICS

- 1 A row-level field can be aggregated later.
- 2 Mixing aggregate and non-aggregate expressions causes an error.
- 3 The view grain determines the final aggregation.

DECISION

- 1 Decide whether the business rule applies per record or per displayed group.
- 2 Use consistent aggregation inside the expression.
- 3 Validate with a known small example.

Calculation Function Families

1

Number

ROUND, ABS, CEILING and mathematical operations.

2

String

LEFT, SPLIT, CONTAINS and case functions.

3

Date

DATETRUNC, DATEPART, DATEDIFF and TODAY.

4

Logical

IF, IIF, CASE, AND and OR.

5

Aggregate

SUM, AVG, COUNTD, MIN and MAX.

6

Table calculation

LOOKUP, WINDOW_SUM, RANK and running functions.

IF, IIF and CASE

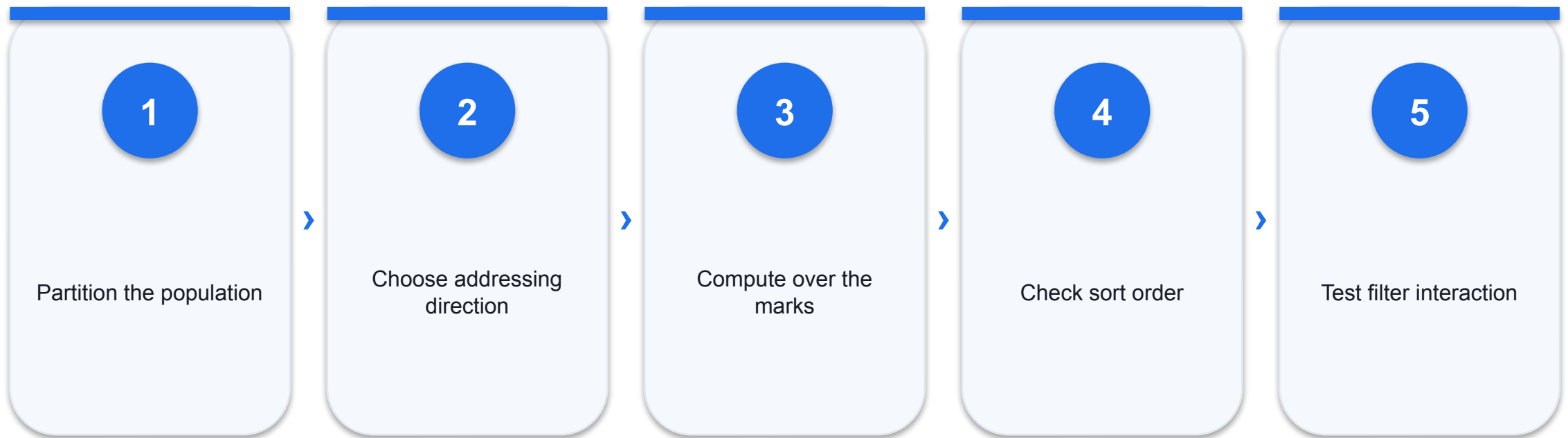
SYNTAX

- 1 IF supports multiple conditions and branches.
- 2 IIF is concise for one true/false decision.
- 3 CASE compares one expression to listed values.

CONTROL

- 1 Keep thresholds in parameters or governed fields.
- 2 Always include an explicit else path.
- 3 Test blanks, zero and boundary cases.

Table Calculation Addressing



Parameter Anatomy

1

Name

Business-facing description.

2

Data type

Integer, real, string, Boolean, date or date-time.

3

Allowable values

All, list or range.

4

Current value

Single state used by calculations or filters.

5

Control

Visible interface for user input.

6

Action

Dashboard marks can update it.

Threshold Parameter Pattern

PATTERN

- 1 User chooses a materiality threshold.
- 2 Calculation compares aggregated performance to the parameter.
- 3 Filter retains the TRUE result.

GOVERNANCE

- 1 Show the current threshold in title or caption.
- 2 Choose a defensible default and range.
- 3 Avoid implying the threshold is data-driven when it is policy.

Dynamic Metric Pattern

Parameter

- Sales, Profit, Margin or Orders
- One selected value

CASE calculation

- Returns the matching measure
- Uses consistent aggregation

Dynamic labels

- Title and tooltip state the metric
- Formatting matches unit

Useful Table Calculations

1

Percent of total

Contribution within the partition.

2

Running total

Cumulative performance over an ordered dimension.

3

Difference from

Change against prior mark or benchmark.

4

Rank

Relative position under the current filter state.

Reference Lines and Parameters

CONTEXT

- 1 Reference lines provide average, median, constant or computed context.
- 2 Parameters can supply a user-controlled benchmark.
- 3 Bands express an acceptable range.

CAUTION

- 1 Label what the reference means.
- 2 Avoid treating an average as a target.
- 3 Keep the scale and filter state visible.

Calculation Validation

1

Small sample

Manually verify several records or marks.

2

Control total

Reconcile numerator, denominator and aggregate.

3

Edge cases

Test zero, null, negative and threshold boundaries.

4

Filter state

Check whether calculation order changes the result.

5

Formatting

Match percentages, currency and units.

6

Explanation

A reviewer can restate the rule in plain language.

A dynamic view still needs a stable definition.

Parameters make analysis interactive; they do not replace governed metric logic and visible assumptions.

Create Dynamic Profitability Analysis

LAB 9

Finance wants to switch between sales, profit and margin, and test which products exceed a user-defined threshold.

EXPECTED OUTPUT

A dynamic metric view with calculated fields, parameter controls, table calculations and a threshold filter.

TOOLS Calculated fields · IF/CASE · table calculations · parameters · dynamic titles

Lab 9 Acceptance Evidence

ACCEPTANCE TEST

Metric and threshold controls change the view correctly, formulas reconcile to Excel controls and invalid denominator cases are handled.

Build

Packaged Tableau workbook
Required views and controls

Prove

Screenshot with filter state
Excel reconciliation check

Explain

Observation and meaning
Limitation and next action

Recap — Calculations and Parameters

1

Knowledge

Row-level calculation and related principles.

2

Ability

K2 · A5 · LO5

3

Evidence

A packaged Tableau workbook plus reconciliation and interpretation.

4

Next

Connect this layer to the next decision requirement.

TOPIC 06

Analytics and Communication

Reference lines · trends · forecasts · clusters · limitations · data stories

Key Concepts — Analytics and Communication

1

Reference context

Average, median, target and distribution bands help the audience judge performance.

2

Trend models

A fitted line summarises association; it does not establish causality.

3

Forecast

Tableau extends a time series under model assumptions and expresses uncertainty.

4

Clustering

Groups similar marks from selected features; business labels require interpretation.

5

Limitations

Data coverage, quality, granularity, bias and model assumptions bound the conclusion.

6

Responsible narrative

Separate observation, interpretation, recommendation and uncertainty.

Analytics Pane: Context and Models

1

Reference line

Constant, average, median or computed benchmark.

2

Reference band

Range between values or distribution limits.

3

Trend line

Fitted relationship across marks.

4

Forecast

Extension of a time series with uncertainty.

5

Cluster

Algorithmic grouping from selected measures.

6

Box plot

Distribution summary and unusual marks.

Average Line versus Target Line

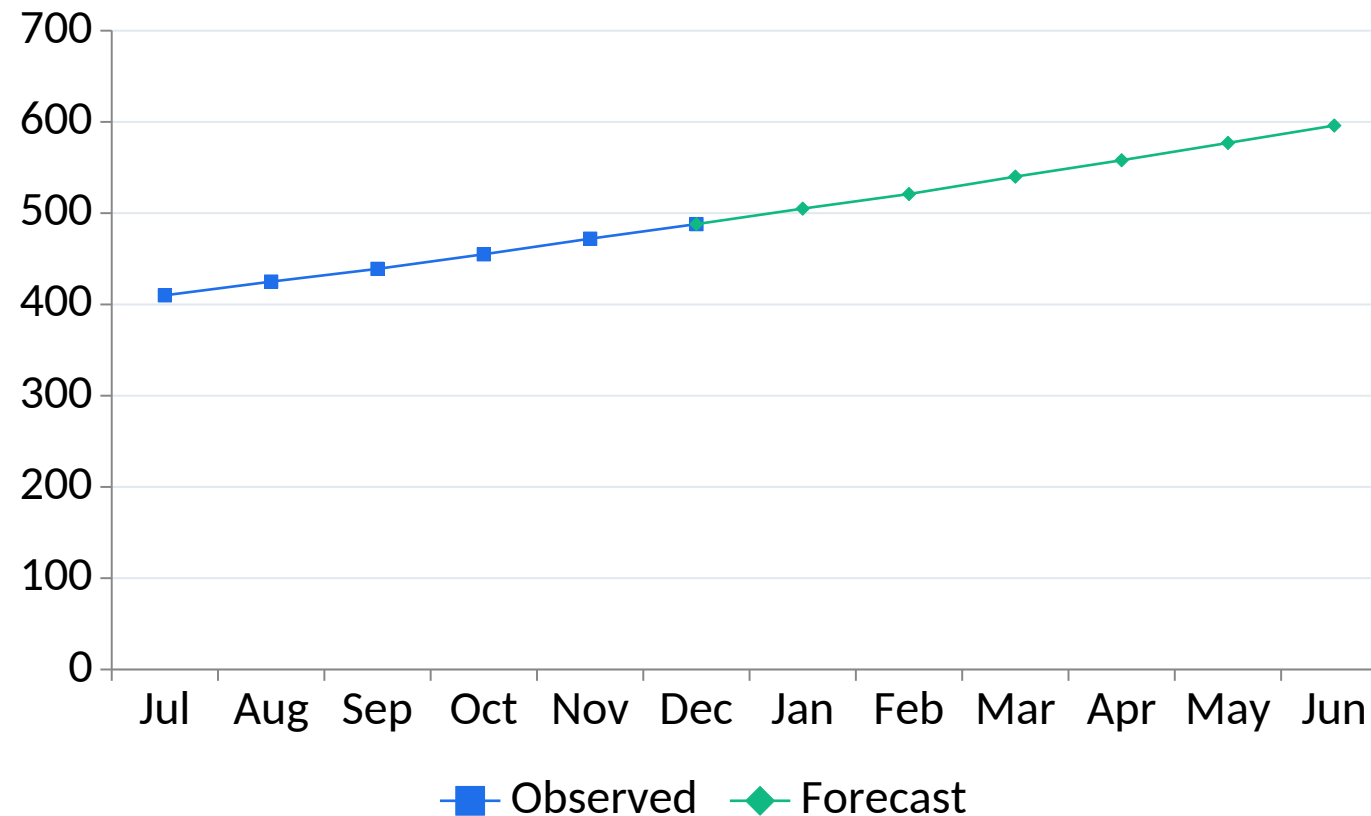
OBSERVED

- 1 Average summarises the observed population.
- 2 It moves when filters or the population change.
- 3 Useful for relative comparison.

EXPECTED

- 1 Target represents policy, plan or benchmark.
- 2 It should have an owner, period and source.
- 3 Do not label an average as expected performance.

Observed Trend and Forecast Context



READ THE EVIDENCE

The forecast extends the recent pattern, but the decision should use its prediction interval and monitor error after each refresh.

Editable chart: change the embedded data, labels, axes and colours directly in PowerPoint.

Trend Lines

EVIDENCE

- 1 Describe direction and fitted relationship.
- 2 Model form and level of detail affect the line.
- 3 R-squared summarises fit in the displayed sample.

LIMIT

- 1 Association is not causation.
- 2 Outliers and clusters can dominate the fit.
- 3 Inspect residual patterns and domain plausibility.

Forecast Design

1

Time grain

Consistent intervals with enough history.

2

Horizon

Short enough for assumptions to remain plausible.

3

Seasonality

Recurring pattern supported by complete cycles.

4

Prediction interval

Range of uncertainty, not decorative shading.

5

Missing values

Treatment can change the model.

6

Monitoring

Compare forecast with actuals and refresh.

Clustering Workflow

Select features

- Decision-relevant measures
- Comparable scaling and meaning

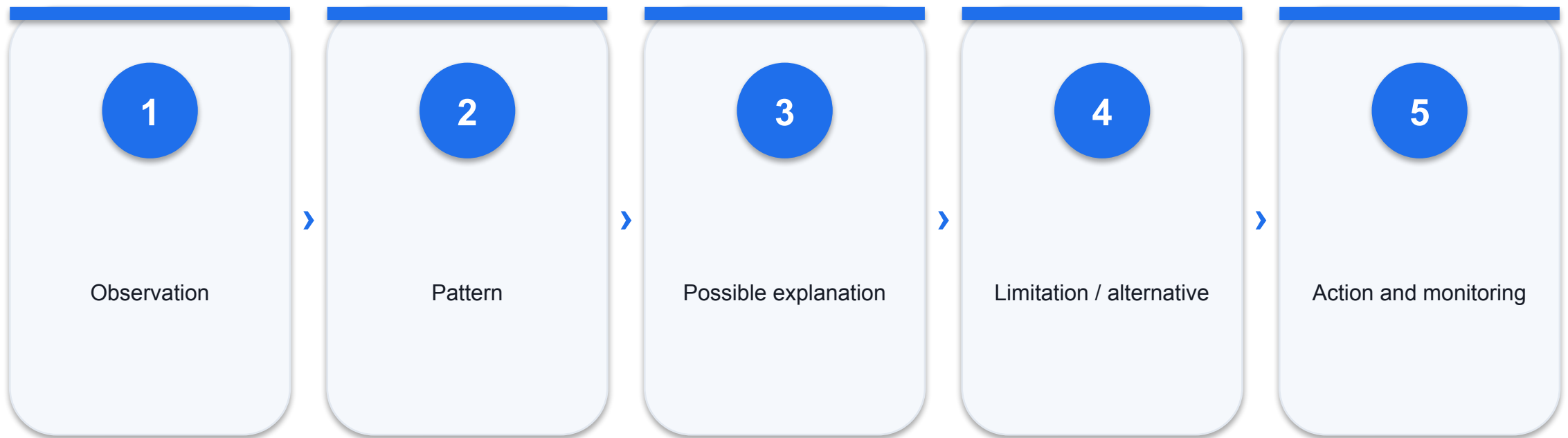
Inspect profiles

- Cluster centres and membership
- Outliers and stability

Name cautiously

- Business labels after review
- Do not stereotype individuals

Interpretation Ladder



Common Data Limitations

1

Coverage

Period, channel or population may be incomplete.

2

Quality

Missing, duplicated or inconsistent values.

3

Granularity

Aggregation can conceal within-group variation.

4

Bias

Selection or measurement process shapes the result.

5

Drift

Historical relationships may change.

6

Confounding

A third factor may explain an apparent relationship.

Correlation and Causation

CAN SAY

- 1 Correlation describes co-movement.
- 2 Trend lines estimate association under assumptions.
- 3 Useful for exploration and prediction.

CANNOT CLAIM

- 1 Causation needs design and alternative explanations.
- 2 Reverse causality and confounding remain possible.
- 3 Use cautious language in recommendations.

Responsible Data Story

Evidence

- What the visual directly shows
- Population, period and unit

Interpretation

- Most plausible business meaning
- Alternative explanations

Action

- Owner, trigger and next check
- Uncertainty and guardrail

Analytical Review Questions

1

Right question

Does the view address a real decision?

2

Right population

Do filters and exclusions match the claim?

3

Right method

Is the visual/model suitable for the data?

4

Right context

Are benchmarks, units and time visible?

5

Right caveat

Are material limitations explicit?

6

Right action

Can someone act and monitor the outcome?

The strongest visual is honest about what it cannot prove.

Communicate the signal, the uncertainty and the next evidence needed for a better decision.

Analyse Trends, Forecasts and Clusters

LAB 10

The strategy team needs a cautious outlook for outlet demand and a segmentation of store operating patterns.

EXPECTED OUTPUT

An analytics workbook with reference context, trend, forecast, clustering and a limitations-led decision narrative.

TOOLS Reference lines · trend lines · forecast · clustering · describe trend/model

Lab 10 Acceptance Evidence

ACCEPTANCE TEST

The workbook states assumptions and limitations, distinguishes association from causation, and reports at least one action with a monitoring trigger.

Build

Packaged Tableau workbook
Required views and controls

Prove

Screenshot with filter state
Excel reconciliation check

Explain

Observation and meaning
Limitation and next action

Recap — Analytics and Communication

1

Knowledge

Reference context and related principles.

2

Ability

K1 · A6 · LO6

3

Evidence

A packaged Tableau workbook plus reconciliation and interpretation.

4

Next

Connect this layer to the next decision requirement.

WRAP-UP

Evidence, Assessment and Next Steps

Turn visual analysis into a defensible decision.

What You Achieved

1

LO1

Apply basic Tableau features for displaying information.

2

LO2

Synthesise various plots to present data visually.

3

LO3

Transform data to create informative and dynamic data displays.

4

LO4

Create dashboards and scorecards to display internal and external benchmark data.

5

LO5

Apply calculations and parameters to create interactive graphics, visuals and technical features.

6

LO6

Perform analytics and communicate data limitations and interpretations of findings.

Courseware and Assessment on the LMS

Courseware

- Learner slide PDF
- Detailed Learner Guide
- Ten individual lab folders

Assessment

- WA question paper
- PP question paper
- Upload your own evidence

Support

- Check the current TGS code
- Use the current release
- Contact trainer or academy

Open LMS / TMS

Tableau Practice Exam

Purpose

- Check concept readiness before assessment
- Use results to target revision

Focus

- Data roles, modelling and chart choice
- Dashboards, calculations and responsible interpretation

Readiness

- Complete individually
- Review explanations, then revisit the relevant topic and lab

[Open practice exam](#)

Assessment Reminder

WA

- 5 open-ended knowledge questions
- 60 minutes

PP

- 6 connected Tableau tasks
- 60 minutes

Submission

- Complete assessment attendance
- Upload papers and required evidence

Assessment Flow



Digital Attendance (Mandatory)

1

Assessment

Scan the SSG QR code before starting.

2

Training

Confirm AM and PM records for both days.

3

Minimum

At least 75% attendance is required.

4

Support

Tell the trainer immediately if attendance submission fails.

THANK YOU

Make the evidence easy to see — and hard to misunderstand.

Connect, model, visualise, interact, analyse and communicate with explicit limits.